

Multiple Choice (10 marks)

1. Besides temperature, what other physical conditions must be taken into account for the growth of bacteria?
 - (a) Levels of oxygen, acidity or alkalinity of the medium
 - (b) Temperature changes, light exposure
 - (c) Atmospheric pressure, noise levels
 - (d) Nitrogen levels, radiation exposure
 - (e) Soil texture, light wavelength
2. Two individuals, one with type B blood and one with type AB blood, have a child. The probability that the child has type O blood is
 - (a) 50%
 - (b) 100%
 - (c) 0%
 - (d) 10%
 - (e) 75%
3. What functions are served by sound communication in animals?
 - (a) Mate attraction, species recognition, defense, and territorial claim
 - (b) For heat regulation
 - (c) For hunting
 - (d) For navigation
 - (e) For mineral absorption
4. Targeting of a newly synthesized protein is most likely to require two different signal peptides for which of the following destinations?
 - (a) Ribosome
 - (b) Mitochondrion
 - (c) Chloroplast
 - (d) Lysosome
 - (e) Endoplasmic reticulum
5. The amino acid ornithine is regenerated during each turn of the urea cycle in a similar manner to which of the following participants in the citric acid cycle?

- (a) Alpha-ketoglutarate
- (b) Glutamate
- (c) Acetyl CoA
- (d) Oxaloacetate
- (e) Citrate

True / False (10 marks)

Answer True or False

6. True or False: Menopausal hormone therapy and irregular endometrial bleeding: a potential role for uterine natural killer cells?
7. True or False: Dose end-tidal carbon dioxide measurement correlate with arterial carbon dioxide in extremely low birth weight infants in the first week of life?
8. True or False: Do people who were passive smokers during childhood have increased risk of long-term work disability?
9. True or False: Profiling quality of care: Is there a role for peer review?
10. True or False: Can third trimester ultrasound predict the presentation of the first twin at delivery?

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the various types of evidence from living organisms that support the theory of evolution, analyzing their significance and interconnections in the evolutionary framework.
12. Discuss the processes of leaching and nutrient cycling in tropical rain forest soils, and analyze how these processes impact ecosystem dynamics and biodiversity within this biome.

End of Paper